

1 · METRICS & ALGORITHMS

House Price Prediction	Regression metrics → MSE / RMSE / MAE
Data Imbalance (tabular)	Optimize F1-score + apply SMOTE
Data Imbalance (images)	Optimize F1-score + augment new images
Tabular Data (default)	LightGBM (~90% of cases)
Non-linear + interdependent	XGBoost — also handles imbalanced data
High-dimensional sparse data	Factorization Machines
Document topic modelling	LDA (Latent Dirichlet Allocation)
RCF (Random Cut Forest)	Unsupervised anomaly detection — NOT for binary. Keywords: unplanned events, unusual patterns.

■ *L1 shrinks coefficients to zero → feature selection. L2 penalizes large weights → generalization. Early stopping is separate.*

2 · SAGEMAKER TRAINING

Custom script + SageMaker	SageMaker Script Mode (~90% of cases)
Large model > 1 GPU	FSDP — splits model across GPUs
Increase no. of instances	SMDDP (SageMaker Distributed Data Parallel)
Compute bottleneck	SMDDP library
Memory bottleneck	Spark on EMR
DDP + Python script	SageMaker Profiler annotations & reports
Hyperparameter tuning	Bayesian (uses past trials) + Hyperband (early stopping)
Will more data help? (SGD)	Use Learning Curve to decide
Opt-out training metadata	CLI/SDK → env var opt_out_tracking=1. SageMaker UI → disable manually.

3 · SAGEMAKER INFERENCE

Same framework, multiple models	Multi-Model Endpoint (MME)
Different frameworks	Multi-Container Endpoint
Pre/infer/post in one call	Multi-Container Endpoint (serial pipeline)
Once daily, small payload	Serverless Inference — MaxConcurrency=1
Serverless, no cold start	Provisioned Concurrency
Cold start monitoring	CloudWatch metric: ModelSetupTime
Default auto-scaling metric	InvocationsPerInstance
Variant traffic split	Weights are relative — A=3, B=1 → 75%/25%
Cost: multiple models	Multi-Model Endpoint on same endpoint
Model compiler (normal)	SageMaker Neo
Big / deep learning models	Deep Learning Containers (DLC)

4 · SAGEMAKER STUDIO & NOTEBOOKS

Lifecycle Configuration	Auto-install packages, clone repos, env vars, mount EFS, auto-stop idle
Debugging GPU/CPU/gradients	SageMaker Debugger → TensorBoard for visualisation
Pause pipeline for approval	SageMaker Callback Step
Model versioning	SageMaker Model Registry (auto-increments versions)
Organise models into groups	SageMaker Model Registry Collections
Explainability	Simple model (Linear Learner) for explainability; complex for accuracy
Feature Store sync	Spark Connector (offline + online)
Add feature to feature group	UpdateFeatureGroup + PutRecord commands
Feature Store setup order	1) Create Feature Group → 2) Ingest records → 3) Access store
Performance after dataset	1) Tune hyperparams → 2) Evaluate metrics → 3) Provision Throughput

5 · DATA & STORAGE

Preparing data for ML	AWS Data Wrangler (99%)
Non-ML ETL / data pipelines	AWS Glue
Streaming data ingestion	Kinesis Firehose (only when explicitly stated)
ETL slow / throttled	Glue quota reached → request quota increase
Import data into Data Wrangler	Cataloged Connection
Semantic search on S3 docs	Kendra S3 Connector (ingest) + Kendra (query)
Low-effort RAG	Amazon Kendra Experience Builder
Detect PII (S3)	Amazon Macie
Remove PII	AWS Glue DataBrew
Detect PHI	Amazon Comprehend Medical
Network file storage (snapshots)	Amazon FSx for NetApp ONTAP
Large-scale ML training data	FSx for Lustre — links directly to S3, high throughput
Share data without transforming	Dynamic Data Masking
Automated labeling (no humans)	SageMaker Ground Truth automated labeling

6 · NETWORKING & SECURITY

Bedrock from private VPC/EC2	PrivateLink interface endpoint — do NOT modify outbound rules
Cross-account S3 (no public IP)	S3 Gateway Endpoint + VPC Interface Endpoints + update bucket policy
Comprehend model cross-account	ImportModel API (same-region, A → B)
S3 in VPC	S3 cannot be placed in VPC — use PrivateLink / Gateway Endpoint
Encrypt inter-container traffic	Enable intercontainer traffic encryption setting
Rotate API keys on schedule	AWS Secrets Manager + Lambda
Block specific IP	Network ACL (stateless, supports explicit deny by IP)
Data lake access control (efficient)	Lake Formation — IAM on vast unstructured data is unmanageable
Network isolation (training)	Enable Network Isolation → blocks all internet from container

7 · MLOPS & ORCHESTRATION

General / ML pipelines	Amazon SageMaker Pipelines
Multi-service orchestration	AWS Step Functions (Lambda, Glue, etc.)
Complex DAGs	Amazon MWAA (Managed Airflow) — rarely tested
CloudFormation: host a model	AWS::SageMaker::Model resource
Rolling deployment	Gradually replaces instances; batch size = speed vs safety
Blue/Green deployment	Switch all traffic at once after validation. Easy rollback.
Canary deployment	Small % to new version first, then expand
Inefficient resource usage	AWS Compute Optimizer
Notebook auto-config	SageMaker Lifecycle Configuration
LLM inference optimisation	Speculative Decoding
Bedrock fine-tuning (no infra mgmt)	Amazon Bedrock fine-tuning jobs

8 · CHARTS & VISUALISATION

Chart	Use for	Example
Box Plot	Outliers, distribution comparison	Unusually high spending
Scatter Plot	Outliers, correlations	Age vs income unusual points
Histogram	Distribution & skewness	Is age normally distributed?
Vertical Bar Chart	Category counts, entity frequency over time	Customers by country
Line Chart	Trends over time	Daily sales over a year
Heatmap	Feature correlations	Income vs age vs spending
Pie Chart	Percentage of a whole	Market share by plan
Density Plot	Compare distributions of groups	Churned vs non-churned

Feature Splitting	Extract parts from a value — e.g. timestamp → year/month/day
Binning	Group continuous → buckets. E.g. age → age range
Log Transformation	Reduce skewness — e.g. square footage
One-Hot Encoding	Unordered categories — e.g. neighbourhood
Standardization	Center at 0, $\sigma=1$. Use before distance-based algorithms
Non-linear correlation	Use Spearman — Pearson assumes linearity
Step Scaling	Adds/removes resources in varying amounts as metric crosses thresholds

9 · ANOMALY DETECTION — THRESHOLD CHANGES

Change	Anomaly Detection Freq.	Recall
Lower threshold decreased	Decreased	Decreased
Lower threshold increased	Increased	Increased
Upper threshold increased	Decreased	Decreased
Upper threshold decreased	Increased	Increased

10 · MULTI-MODEL ENDPOINT — DEPLOYMENT STEPS

Step	Action
1	Upload existing models to the same S3 bucket path.
2	Create a SageMaker model with multi-model endpoints enabled.
3	Create endpoint config referencing a multi-model container.
4	Deploy real-time inference endpoint using the config.
5	Update models to use the new endpoint ID; pass model IDs.

11 · MODEL MONITOR — 4 DRIFT TYPES

Monitor Type	What it detects	Trigger keywords
Data Quality	Input feature drift — nulls, ranges, schema changes	feature drift, schema, data quality
Model Quality	Accuracy/metric drift. Needs ground truth labels.	accuracy drop, F1/RMSE, ground truth
Bias Drift (ModelBiasMonitor)	Fairness drift across demographic groups	bias, fairness, demographic parity
Feature Attribution (ModelExplainability)	SHAP values changing in production	SHAP, feature attribution, explainability

■ Clarify = quantify bias + SHAP pre/post training. Model Monitor = scheduled drift detection on deployed endpoints. Glue DataBrew is WRONG for bias.

12 · MONITORING & OBSERVABILITY TOOLS

CloudWatch	Performance metrics, alarms, logs, dashboards. Use for latency, 5XX, CPU/GPU.
CloudTrail	Audit — WHO called WHICH API and WHEN. Trigger: CloudTrail → EventBridge → Pipeline.
AWS Config	Tracks resource configuration changes. Compliance rules.
X-Ray	Distributed tracing for Lambda/API Gateway. Does NOT natively trace SageMaker InvokeEndpoint.
SageMaker Debugger	GPU/CPU/gradient monitoring. Built-in rules for vanishing gradients, overfitting.
CloudTrail log integrity	Enable Log File Validation → signed digest files detect tampering.
emit_metrics	Use in Data Quality Monitor to emit per-feature stats (mean, std, quantiles) to CloudWatch.

13 · INFERENCE MODE — DECISION TABLE

Mode	Use when	Key limits / notes
Batch Transform	High-volume offline, S3 in/out	Cheapest. No persistent endpoint.
Async Inference	Large payload, long processing, queue is OK	Up to 1 GB / 1 hr. Scales to ZERO. Not for real-time fraud.
Serverless	Spiky/unpredictable traffic, low sustained load	Has cold start. Provisioned = no cold start. No GPU support.
Real-time	Sub-second latency, always-on	Up to 5 MB. Use for fraud, banking, IoT.

■ Human waiting → Real-time | Nobody waiting → Batch Transform | Sporadic → Serverless | Large payload → Async

14 · DEPLOYMENT STRATEGIES & EDGE

Shadow Testing	New model gets same live traffic but results NOT served to users. Zero user impact.
A/B Testing	Both models serve real users. Use variant weights to split traffic.
SageMaker Neo	Compile once, run anywhere — cloud AND edge devices.
Edge Manager	Fleet management for edge devices. OTA model updates. Keywords: IoT, offline inference.
Spot Instances	Training & batch transform only. NOT real-time endpoints (interruptions unacceptable).
Scheduled Scaling	Traffic spike known in advance → use Scheduled Scaling first.
Retraining trigger	CloudTrail logs Monitor violation → EventBridge → SageMaker Pipeline retraining.

15 · COST OPTIMIZATION

Compute Optimizer	What should I resize? (right-sizes EC2)
Cost Explorer	Where is my money going? (historical analysis)
AWS Budgets	Warn me before I spend too much. (threshold alerts)
Trusted Advisor	Flags underutilised/idle resources.
Inference Recommender	Right-sizes instance type for ML endpoints specifically.
Async scales to zero	Auto-scales to 0 when idle — major saving for non-sustained workloads.
Spot Training	Up to 90% cost saving with managed spot + checkpoints.
Resource Tagging → Billing	1) Create key-value tags. 2) Attach to resources. 3) Enable Cost Allocation Tags in Billing.

16 · GAPS & ADDITIONS — DOMAINS 1, 2 & 3

Kinesis Streams vs Firehose	Streams = real-time processing + custom consumers. Firehose = delivery to S3/Redshift/OpenSearch.
AUC-ROC vs PR Curve	Use PR curve when class imbalance is severe. AUC-ROC can mislead on minority class.
SageMaker Autopilot	AutoML — auto algorithm selection + HPO. Keywords: automatic, no ML expertise.
SageMaker JumpStart	Pre-built models hub. Keywords: foundation model, one-click deploy, no training.
SageMaker Experiments	Tracks training run metadata. NOT deployment governance (that's Model Registry).
SageMaker Lineage Tracking	Tracks data + model artifact lineage. NOT for enforcing promotion policies.
Real-time + unpredictable burst	Real-time endpoint + Application Auto Scaling. Spot Fleet behind ALB = WRONG.
Bedrock Guardrails	Filters harmful content, PII, off-topic. Separate from IAM/VPC security.
Glue DataBrew vs Wrangler	DataBrew = visual point-and-click cleaning. Wrangler = within SageMaker Studio.

17 · SECURITY, ENCRYPTION & ADDITIONAL PATTERNS

SSE-S3	AWS-managed key. No extra cost. Not for compliance requiring customer key control.
SSE-KMS	KMS key (AWS-managed or CMK). Auditable via CloudTrail. Slower than SSE-S3.
SSE-C	Customer-provided key per request. Highest control. AWS never stores the key.
Client-Side Encryption	Encrypt before sending to S3. AWS Encryption SDK. S3 never sees plaintext.
KMS Key Policies	JSON policies — who can use the key. Cross-account access, resource scoping.
Centralized IAM notebooks	One shared IAM role → all notebook instances. Reduces management overhead.
Inter-node encryption	Enable 'encrypt inter-node training communication' in job config.
Customer-managed KMS + SDK	For application-level key control on S3 training data. SSE-KMS (AWS-managed) insufficient.

18 · ADDITIONAL PATTERNS

Serverless — GPU	Serverless inference does NOT support GPU. Use real-time endpoint if GPU needed.
Minimize false positives	Increase Precision. Minimize false negatives → increase Recall.
Fraud detection — algorithm	Linear Learner (fast, interpretable, handles imbalance).
Non-linear correlation	Spearman correlation. Pearson assumes linearity.
JSONL format	Unstructured + real-time streaming data format.
Large data performance	Filter → Compress → Convert to columnar (Parquet/ORC).
Stacking vs Bagging vs Boosting	Stacking = different algos combined. Bagging = same algo parallel (RF). Boosting = sequential (XGBoost).
SHAP vs PDP	SHAP = local (single prediction). PDP = global (feature effect across dataset).
TensorBoard vs Clarify	TensorBoard = CNN training issues, gradient viz. Clarify = bias, SHAP, prediction explanation.
Provisioned + Auto Scaling	Provisioned handles baseline; Auto Scaling handles bursts.
Neural networks in MCQs	AWS treats neural networks as supervised algorithms — check MCQs carefully.

19 · SAGEMAKER BUILT-IN ALGORITHMS

Algorithm	Type	Use case
K-Means	Unsupervised	Customer segmentation, clustering
PCA	Unsupervised	Dimensionality reduction, visualization
RCF	Unsupervised	Anomaly detection, unusual patterns
LDA	Unsupervised	Document topic modelling
DeepAR	Supervised	Time series forecasting with confidence intervals
BlazingText	Supervised	Fast text classification + word embeddings
NTM	Unsupervised	Neural topic modelling — topic mixtures per doc
Linear Learner	Supervised	Regression, binary/multi-class, fraud detection
XGBoost	Supervised	Tabular, non-linear, imbalanced, tree boosting
Factorization Machines	Supervised	Sparse high-dim data, CTR, user-item recommendations
Image Classification	Supervised	Transfer learning on custom images (ResNet)
Object Detection	Supervised	Bounding boxes. Pre-trained on COCO.
Semantic Segmentation	Supervised	Pixel-level classification. Medical imaging, satellite.
Neural Networks	Supervised	Treated as supervised by AWS in MCQ context.

20 · EVALUATION METRICS

Metric	When to use	Note
Accuracy	Balanced dataset, general classification	Misleading for imbalanced data
Precision	Minimize false positives (spam, search)	$TP / (TP+FP)$
Recall	Minimize false negatives (medical, fraud)	$TP / (TP+FN)$
F1-Score	Balance precision & recall, imbalanced data	Harmonic mean of P & R
ROC-AUC	Threshold-independent, balanced classes	Area under TPR vs FPR
PR-AUC	Imbalanced data — better than ROC-AUC	Area under Precision-Recall curve
Specificity	True negative rate (when negatives matter)	$TN / (TN+FP)$
NDCG	Ranking / search / recommendations	Normalized Discounted Cumulative Gain
Confusion Matrix	Binary classification detail	Shows TP, TN, FP, FN

21 · DRIFT TYPES

Drift Type	Situation	Fix
Concept Drift	Customer behaviour / fraud patterns changed	Retrain with new data
Label Drift	Positive/negative label ratio changed	Adjust thresholds or retrain
Training-Serving Skew	Input features at inference $\neq$ training data	Fix pipeline / preprocessing
Covariate Shift	Feature distribution changes, target relation same	Importance weighting

■ *Meaning changed* → *Concept Drift* | *Distribution changed* → *Data/Label Drift* | *Pipeline mismatch* → *Training-Serving Skew*

22 · FEATURE ENGINEERING

Normalization	Scale to [0,1] or $\mu=0, \sigma=1$. Use before distance-based algorithms.
Missing Value Imputation	Mean/median/forward-fill. Data Wrangler has built-in recipes.
Target Encoding	Replace category with mean target value. High-cardinality solution.
Ordinal Encoding	Ordered categories: low→1, med→2, high→3. Order matters.
Data Leakage	Future info in training. Remove temporal features. Split before preprocessing.
Temporal Features	Extract day, month, year, weekday, hour. Critical for time series.
Interaction Features	Combine two features (age × income). Linear models need this explicitly.

23 · AMAZON BEDROCK — GENERATIVE AI

RAG	Ground LLM responses in company docs. Reduces hallucinations.
Knowledge Bases	Store embeddings in vector DB (Aurora, OpenSearch). Retrieved at inference.
Agents	LLM calls tools/APIs on your behalf (Lambda, databases, external APIs).
Embeddings	Text → vectors. Bedrock Titan Embeddings or third-party models.
Fine-tuning	Adapt model to downstream task with small labelled dataset (~100+ examples).
Continued Pretraining	More raw unsupervised data to improve general domain capabilities.
Bedrock Guardrails	Filter harmful content, PII, off-topic responses. Pre/post model output.

24 · DATA SERVICES — WHICH TO USE

Service	Best for	NOT for
Athena	Ad-hoc SQL on S3. Serverless, pay-per-query.	Real-time analytics.
Glue	ETL pipelines, schema discovery (Crawler).	Real-time. One-off queries.
EMR	Big data Spark/Hadoop, custom code.	Simple pipelines → use Glue.
Redshift	Data warehouse. Fast MPP SQL. BI dashboards.	Unstructured data, streaming.
Kinesis Streams	Real-time ingestion + custom consumer logic.	Delivery to S3 → use Firehose.
Firehose	Streaming delivery to S3/Redshift/OpenSearch.	Processing logic → use Streams.
MSK (Kafka)	High-throughput event streaming, durable queues.	Simple S3 delivery → Firehose.

25 · STORAGE FORMATS

Format	Use for	Note
Parquet	Analytics, ML training	Columnar, fast reads, compressed. Default for ML.
JSON/JSONL	Streaming logs, flexible schema	JSONL = one JSON object per line, ideal for streaming.
XZ	Maximum compression	Slowest. Use for archival / storage cost optimization.
Snappy	Fast reads, low CPU overhead	Balanced. Often used with Parquet.
Gzip	Balanced compression + speed	Good general-purpose. Compatible with most tools.

26 · BIAS METRICS — SAGEMAKER CLARIFY

Metric	What it checks	Memory trick
CI	Are demographic groups equally represented?	Do I have enough examples of each group?
DPL	Do positive outcomes occur at same rate across groups?	Do groups get positive outcomes equally?
CDD	Does outcome differ after controlling for relevant factors?	After controlling for factors, is bias still present?
TVD	How different are two probability distributions overall?	How different are the distributions?
KL Div.	How much has production distribution drifted from training?	How much has distribution drifted?

27 · COST TOOLS — ONE-LINER RULES

Compute Optimizer	What should I resize? (right-size EC2 / ML instances)
Cost Explorer	Where is my money going? (historical spending analysis)
AWS Budgets	Warn me before I spend too much. (threshold + alerts)

28 · SCENARIO → METRIC

Scenario	Metric
Medical / cancer / disease detection	Recall
Fraud detection	Recall
Defect detection in manufacturing	Recall
Spam filtering	Precision
Search results	Precision, NDCG
Recommendations	CTR, Precision@K, NDCG
Imbalanced classification	F1, PR-AUC
Balanced classification	Accuracy
Ranking problems	NDCG

■ Missing positives is expensive → Recall | False alarms are expensive → Precision

29 · SCENARIO → ALGORITHM

Scenario	Algorithm
Binary classification	XGBoost, Linear Learner
Regression	Linear Learner
Customer segmentation	K-Means
Topic discovery	NTM
Recommendation (user-item interactions)	Factorization Machines
Forecasting / time series	DeepAR
Text classification	BlazingText
Image classification	Image Classification (SageMaker built-in)
Object localization (bounding boxes)	Object Detection
Pixel-level image labeling	Semantic Segmentation
Anomaly detection (unsupervised)	RCF
High-dim sparse data / CTR / recommendations	Factorization Machines

Inference by Requirement:

Requirement	Answer
Low latency, always available	Real-time Endpoint
Large payload / long processing	Async Inference (up to 1 GB, 1 hr)
Sporadic / spiky traffic	Serverless Inference
Variable sustained traffic	Real-time + Application Auto Scaling
Large offline batch	Batch Transform
Once daily, small payload	Serverless (MaxConcurrency=1)
■ <i>Human waiting</i> → <i>Real-time</i> <i>Nobody waiting</i> → <i>Batch</i> <i>Sporadic</i> → <i>Serverless</i> <i>Burst known</i> → <i>Scheduled Scaling</i>	

Training by Requirement:

Requirement	Answer
Distribute data, single huge model	Data Parallelism (SMDDP)
Model too large for one GPU	Model Parallelism (FSDP)
Hyperparameter search	SageMaker HPO — Bayesian + Hyperband
Reduce training cost	Managed Spot Training (up to 90% saving)
Detect gradients / overfitting	SageMaker Debugger built-in rules

Explainability by Requirement:

Requirement	Answer
Feature importance / SHAP	SageMaker Clarify
Explain individual prediction	SageMaker Clarify
Bias detection pre/post training	SageMaker Clarify
Visualise gradients / loss (CNN)	TensorBoard + SageMaker Debugger
Simple interpretable model	Linear Learner

31 · EXAM GOLD NUGGETS

Recommendation + clicks/views/purchases	→ Factorization Machines
Time series forecasting	→ DeepAR
Customer segmentation	→ K-Means
Text classification	→ BlazingText
Topic extraction	→ NTM
Need highest recall	→ Medical / Fraud use cases
Need highest precision	→ Spam / Search use cases
Real-time prediction	→ SageMaker Real-time Endpoint
Offline prediction — millions of records	→ Batch Transform
ETL on S3	→ AWS Glue
Interactive SQL on S3	→ Amazon Athena
Spark mentioned	→ EMR
Storage cost optimization	→ Highest compression → XZ
Training cost reduction	→ Managed Spot Training
Interpret predictions / bias	→ SageMaker Clarify
Use company docs with LLM	→ RAG + Bedrock Knowledge Bases
Multi-step LLM workflow	→ Bedrock Agents
Semantic / vector search	→ Embeddings + OpenSearch / Kendra
No-code ML for business users	→ SageMaker Canvas
No-code image / video analysis	→ Amazon Rekognition
No-code time series (managed)	→ Amazon Forecast (vs DeepAR = custom training)
No-code recommendations (managed)	→ Amazon Personalize
L1 regularization	→ Feature selection — shrinks coefficients to zero
L2 regularization	→ Weight regularization — reduces overfitting
Neural network in MCQ	→ Treat as supervised algorithm (AWS exam pattern)